

Ravi Teja Pasupuleti

Building agentic AI systems that turn LLMs into production-grade decision engines.

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PROFESSIONAL EXPERIENCE

Senior AI Engineer — Ngenux Solutions

Jan 2025 - Present

- Contributed to **SeekAlfred** (seekalfred.ai), Ngenux's flagship **AI Chief of Staff** platform for C-suite leaders CMO that unifies fragmented signals across the marketing and revenue stack — **Google Ads**, **Meta**, **LinkedIn Ads**, **HubSpot** — into decision-ready recommendations. Built agentic modules for **Spend-Mix optimization**, **Creative Fatigue Detection**, and an **AI Visibility Score** using **LLMs** orchestrated with **LangGraph**, **PGVector** for retrieval, and **FastAPI** services deployed on **AWS ECS** with **API Gateway**.
- Engineered a **real-time Voice2Voice conversational agent** with sub-second latency using a streaming **STT** → **LLM** → **TTS** pipeline — **Sarvam AI** for transcription, **Claude Sonnet 4.5** for reasoning, and **Sarvam AI** for natural speech synthesis — orchestrated over **LiveKit (WebRTC)** with **VAD-based barge-in** and interruption handling. Containerized with **FastAPI** + **Docker** on **AWS ECS**, with **Redis**-backed session memory and **tool-calling** to backend APIs, enabling natural multi-turn voice dialogue for customer support and lead-qualification workflows.
- Developed a scalable, **event-driven document processing system** on **AWS** to automate document classification and information extraction. Implemented a multi-threaded **SQS listener** on **ECS** for high-throughput request handling. Leveraged **GenAI models** (Gemini) within a multi-engine framework to perform classification and data extraction using dynamic configurations from **AWS Parameter Store** and a **Taxonomy API**. Secured credentials via **Secrets Manager**, stored documents/results in **S3**, and persisted structured outputs in **Aurora DB** for analytics.
- Built an **agentic AI system** for an EdTech client using **LangGraph**, enabling **RAG-based chat**, real-time human handoff, and automated **Salesforce** ticket creation. Integrated **Amazon Bedrock** (Titan, Claude Sonnet) with **OpenSearch** as a vector DB. Deployed Dockerized **AWS Lambda** functions for document ingestion and chatbot interaction, leveraging **S3** for storage and **DynamoDB** for chat history.
- Developed and deployed scalable, containerized backend APIs for **Ngen Studio (AI Studio)** on **AWS EC2** using **FastAPI** and **Docker**. Engineered advanced **agentic AI routing** for enterprise search by integrating **OpenAI Agents**, connecting to **RAG sources** including Google Drive, S3, OneDrive, and document uploads. Enabled **natural language querying** of PostgreSQL, MySQL, MongoDB, and DynamoDB via **Model Content Protocol (MCP)**.
- Architected enterprise **TEXT2SQL solution for Logistics Client** using **AWS Bedrock**, API Gateway, Lambda, S3, Glue, Athena, Neptune, DynamoDB, and PG Vector that transformed natural language to SQL with **80% accuracy**. Delivered **self-correcting solution** that reduced query resolution time, democratizing business intelligence access while maintaining cost efficiency through optimized AWS serverless resource utilization.
- Developed an end-to-end **e-commerce sales call analytics system** utilizing **Whisper** and **NVIDIA NeMo** for speaker diarization, containerized with **Docker** and deployed on **AWS ECS** for scalable processing, integrating **AWS Bedrock** for AI-powered feature extraction and delivering insights through a containerized **Plotly Dash dashboard** with ECS auto-scaling capabilities.
- Engineered an **ML classification system** for EdTech client predicting **student churn**. Implemented streamlined architecture with **S3** for data storage and **SageMaker** for model training and scheduled batch predictions of **5K+ student records** weekly, resulting in a **35% reduction in churn** within one quarter.

Data Scientist — GlobalData Plc

Apr 2023 - Dec 2024

- Assisted in building a **Streamlit-based web application** integrated with a **GPT-powered LangChain agent**. The system enables users to inquire about companies or topics, with GPT formulating a plan passed to the agent, which executes predefined functions to produce detailed outputs, including analyses and data collection from the backend using API. Deployed using **AWS Lambda**, API Gateway, and **S3** for scalability and reliability.

- Fine-tuned **Phi-3.5** for tweet summarization and **BERT** for tweet and Reddit post classification, deploying **Dockerized FastAPI inference services on AWS EKS (Kubernetes)** with **horizontal pod autoscaling and ALB-based load balancing** for scalable and resilient LLM serving.
- I have contributed to the team developing a **daily newsletter** featuring the top 10 trending companies, concepts, and hashtags determined by a **buzz index**. The newsletter includes a summarization of daily tweets related to these trends. Additionally, an automated **Trend Pulse PPT** is generated specifically for companies data.
- Conducted **Selenium web scrapings** and automated **revenue-generating presentations** using Twitter and YouTube data. Quarterly, automated company competitive reports for **3000 companies**, each selling for 100 dollars, resulting good revenue per quarter.

Data Scientist — Tata Consultancy Services (TCS)

Aug 2021 - Apr 2023

- Built an **ML-driven demand forecasting system for a Coca-Cola bottling partner**, modeling **SKU × store × week** sales with **LightGBM** and **LSTM / Temporal Fusion Transformer** ensembles, engineering features from **promotions, weather, holidays, and local events**. Productionized on **AWS SageMaker** with **Glue + Athena** for feature pipelines and **Step Functions** for orchestration; lifted forecast accuracy from **~70% to ~90%** and reduced **stockouts by ~25%**, sharpening production planning and route-level distribution decisions.
- Developed a **customer churn prediction model (XGBoost)** for a retail client, achieving a **15% reduction in churn** and boosting customer lifetime value. The project involved advanced **feature engineering**, model tuning, and **interpretability techniques** to identify at-risk customers and enable targeted retention strategies.
- Developed a **LSTM demand forecasting system** for a retail client, leveraging historical sales, promotions, and external factors to predict weekly product demand with **23% improved accuracy**. Integrated the solution with **inventory optimization**, reducing costs by **15%** and enhancing product availability across many stores.
- Collaborated on **cross-functional teams** to address data-related challenges, leading data analysis for informed decision-making. Utilized advanced **data visualization techniques** like **PowerBI** and **Plotly Dash** for effective communication.

TECHNICAL SKILLS

AI/ML:	Machine Learning, Deep Learning, NLP, Agentic AI, Statistical Modeling
Cloud & MLOps:	AWS (SageMaker, Lambda, EC2, S3, ECS),Palantir ,Docker, Kubernetes, CI/CD Pipelines
Software Engineering:	Python (Advanced), FastAPI, Flask, REST APIs, Microservices Architecture
Data Engineering:	PostgreSQL, Vector Databases (Chroma, Pinecone), PySpark, ETL Pipelines, Power BI,Dagster

EDUCATION

Full Stack Data Science Program	<i>Ineuron</i>
BTech (Electrical & Electronics Engineering)	<i>Jun 2017 - Jun 2021</i>
Sagi Rama Krishnam Raju Engineering College	

CERTIFICATIONS & ACHIEVEMENTS

Azure Data Fundamentals Certificate - Microsoft	View Certificate
DataScientist Certificate - DataCamp	
Problem Solving: Solved 300+ problems on LeetCode and 200+ problems on StrataScratch	View Profiles